JAMA | Original Investigation | CARING FOR THE CRITICALLY ILL PATIENT

4% Tetrasodium EDTA to Prevent Central Venous Access Device–Associated Complications
A Randomized Clinical Trial

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IMPORTANCE Use of central venous access devices (CVADs) can result in catheter-associated bloodstream infection, catheter occlusion, and catheter-related venous thrombosis. EDTA is a potent anticoagulant, antimicrobial, and antibiofilm agent that can be instilled into inactive CVAD ports as a locking fluid to decrease the risk of infection and maintain patency.

OBJECTIVE To determine whether a 4% tetrasodium EDTA (t-EDTA) locking fluid reduces CVAD-associated infections, occlusions, and thrombolytic use in adults treated in an intensive care unit (ICU).

DESIGN, SETTING, AND PARTICIPANTS Pragmatic, triple-blind, multicenter, cluster-randomized crossover trial including patients older than 18 years in the ICU with a CVAD in place and at least 1 lumen not in use. The trial was conducted in 6 Canadian hospitals (3 community hospitals and 3 academic centers). The first patient was enrolled on March 22, 2022, and the date of last follow-up occurred on September 25, 2024.

INTERVENTIONS The ICUs were randomized to use prefilled, identical, masked syringes containing 2.5 mL of t-EDTA (intervention) or control locking fluid (saline or 4% citrate for hemodialysis lines) for all trial participants over a 3.5-month period. After a 1-month follow-up period, ICUs crossed over to the opposing treatment group for another 3.5 months, with a subsequent 1-month follow-up.

MAIN OUTCOMES AND MEASURES The primary outcome was a composite incidence rate of CVAD-associated bloodstream infection, catheter occlusion requiring alteplase use, and/or catheter removal due to occlusion. Three of the 8 prespecified secondary outcomes included incidence rates of each component of the primary outcome separately.

RESULTS Of the 1574 eligible patients, 1468 (mean age, 60 [SD, 16.0] years; 37.7% female) were included in the analysis (696 receiving t-EDTA and 772 receiving control). The incidence rate of primary outcome events was 13.1 per 1000 catheter-days (74 events) in the t-EDTA group vs 19.9 per 1000 catheter-days (126 events) in the control group. In adjusted multivariate analysis, the incidence rate of the composite outcome was lower among patients receiving t-EDTA vs control (rate ratio, 0.68; 95% CI, 0.47-0.96; P = .03). Among the 3 components of the primary outcome, only the secondary outcome of CVAD occlusion requiring administration of alteplase significantly differed between the groups, with lower rates in the t-EDTA group vs control (66 vs 112; incidence rate, 11.67 and 17.73 per 1000 catheter-days, respectively; rate ratio, 0.66; 95% CI, 0.46-0.96).

CONCLUSIONS AND RELEVANCE Compared with control locking solution, use of 4% t-EDTA locking solution for CVADs reduced the incidence of a composite outcome of CVAD-associated bloodstream infection, catheter occlusion requiring alteplase use, and catheter removal due to occlusion among adult patients in the ICU.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: NCT04548713

Visual Abstract
Research Summary
Supplemental content

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JAMA. doi:10.1001/jama.2026.6025
Published online May 18, 2026.

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Use of central venous access devices (CVADs) is common in the intensive care unit (ICU),[1] with up to 80% of adult critical care patients requiring delivery of rapid fluid resuscitation, intravenous antibiotics, total parenteral nutrition, hemodialysis treatment, and vesicant medications such as vasopressors into the central circulation. Types of CVADs most frequently placed in this patient population include nontunneled central venous catheters (approximately 60%), peripherally inserted central catheters (21%), and hemodialysis catheters (14%).[2]

Insertion of a CVAD can result in damage to the blood vessel wall and potentially introduce microorganisms onto internal and extraluminal catheter surfaces.[3] Over time, many bacterial strains have evolved abilities to use the host environment to promote development of their own self-sustaining biofilm colonies.[4-6] Biofilms are a common complication of vascular access devices. In a cohort of 340 adult patients in the ICU, microbiological analysis identified biofilm in 100% of the 359 removed CVADs, regardless of whether clinical infection was present.[7] Studies using scanning electron microscopy and genetic testing have confirmed microorganisms existing within biofilm to have greater rates of antibiotic resistance, tolerance to environmental stressors, and reduced susceptibility to host immune attack compared with planktonic phenotypes (defined as single-celled bacteria that are not attached to biofilm).[8] Presence of biofilm also promotes amplification of the coagulation cascade[6,9] and alterations to clot structure, yielding greater mechanical instability and increased likelihood of dislodgement.[10] Patients with central line-associated bloodstream infection (CLABSI) experience higher mortality rates (24%) and increased ICU lengths of stay longer than 4 days[11-13] and contribute more than $45 000 in health care costs per case compared with patients in the ICU without CLABSI.[14]

Despite efforts to decrease CVAD complication rates,[1] a 2018 meta-analysis of 63 observational and interventional studies involving 50 000 CVADs (396 951 catheter-days) reported CLABSI occurred at a rate of 4.59 per 1000 catheter-days and catheter-associated venous thrombosis at a rate of 8.34 per 1000 catheter-days in adult ICUs.[11] Occlusion affected between 4% and 22% of all CVADs.[11]

A 2023 systematic review of CVAD locking fluids including saline, heparin, citrate, and various ethanol concentrations reported 46.7% citrate to be the only locking fluid associated with reduced central venous catheter and hemodialysis catheter complication rates.[15] However, after a case report of cardiac arrest and death secondary to accidental injection, use of 46.7% citrate locking fluid has been widely discontinued in the US and Canada.[16] Current standard care in adult ICUs includes use of normal saline or heparin as a locking fluid for central venous catheters or peripherally inserted central catheter lines and 4% citrate for dialysis catheters.[17,18] A single-center 1-month observational study of 125 CVADs in 75 patients in an adult critical care setting reported that at least 1 lumen remained locked for an average of 22.4% of the CVAD lifespan over a mean catheterization period of 7 days.[2]

Four percent tetrasodium EDTA (t-EDTA) is a high-affinity metal chelator exhibiting anticoagulant, antimicrobial, and antibiofilm effects due to its ability to sequester divalent cations (calcium, magnesium, and iron) essential for bacterial cell wall integrity and biofilm stability. A 2025 meta-analysis of 8 randomized clinical trials and quasi-experimental and observational studies (n = 403 receiving EDTA; n = 425 receiving a control) revealed that use of EDTA was associated with a significant reduction in catheter-related bloodstream infection or CLABSI (15 vs 71 events; relative risk, 0.18; 95% CI, 0.10-0.33; $P < .001$) and catheter colonization (8 vs 38 events; odds ratio, 0.11; 95% CI, 0.05-0.29) but had no significant effect on use of thrombolytics or all-cause mortality.[14] Results observed during a 24-month implementation of t-EDTA in a sample of 20 adult patients receiving home total parenteral nutrition with previous CVAD complications demonstrated reduced rates of CLABSI (from 1.9 to 0.04 per 1000 catheter-days; $P = .04$) and occlusion (from 11 to 0 cases) compared with the preceding 12-month control period.[19] In a retrospective cohort study of 20 pediatric patients receiving parenteral nutrition, the rate of CLABSI prior to 4% t-EDTA implementation was 2.7 per 1000 catheter-days compared with 0 per 1000 catheter-days during the 12-month intervention period ($P = .002$).[20]

The current trial was designed to test the hypothesis that use of 4% t-EDTA locking fluid would reduce CVAD complication rates compared with current usual care in adult patients in the ICU.

Key Points

Question Can the use of 4% tetrasodium EDTA locking fluid decrease rates of central venous access device (CVAD) complications in adults hospitalized in intensive care units?

Findings In this triple-blind, multicenter, cluster-randomized crossover study of 1468 patients, the incidence of a composite end point of CVAD-associated bloodstream infection, catheter occlusion requiring alteplase use, and catheter removal due to occlusion was 13.1 per 1000 catheter-days (74 events) among patients randomized to receive 4% tetrasodium EDTA locking fluid vs 19.9 per 1000 catheter-days (126 events) in the control group receiving saline or 4% citrate for hemodialysis lines, a statistically significant difference.

Meaning In an adult critical care setting, use of 4% tetrasodium EDTA as a locking fluid reduced a composite outcome of CVAD complications compared with control.

Methods

This study was conducted in accordance with International Council for Harmonization Good Clinical Practice regulations and the ethical principles of the Declaration of Helsinki[21] and followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guideline. The study protocol was approved by harmonized review by the Simon Fraser University, Fraser Health Authority, and Island Health Authority research ethics boards (H20-01458) and the University of Manitoba Biomedical Research Ethics Board (HS25325) and Research Review Committee. All committees approved the use of waiver of consent procedures according to Tri-Council Policy

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Statement 2, Article 3.7A. Further information regarding this consent strategy is provided in eAppendix 1 in Supplement 1.

Trial Population

Eligible patients were older than 18 years, admitted to a critical care unit, and had a CVAD in place with at least 1 lumen not in use. All types of central venous catheters (single, double, triple, and/or quadruple lumens, tunneled and nontunneled), dialysis lines, trialysis catheters, implanted vascular access devices, and peripherally inserted central lines were included. Exclusion criteria included any known sensitivity or allergy to t-EDTA, confirmed or suspected pregnancy, patients who declined to receive blood products, and patients, treating physicians, or temporary substitute decision-makers who declined participation in this study.

Trial Design, Treatments, and Assessments

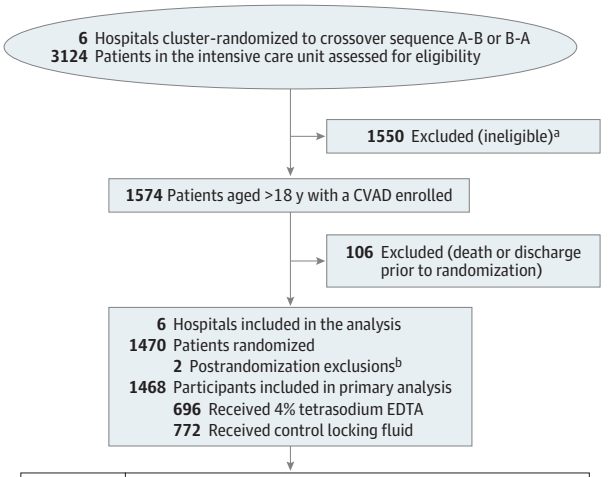
This triple-blind, multicenter, cluster-randomized crossover trial was conducted in 6 Canadian hospitals (3 community hospitals and 3 tertiary care hospitals), including 5 strictly medical-surgical ICUs and 1 ICU that also provided care for patients after cardiac surgery in addition to medical-surgical patients. Additional site characteristics are summarized in eTable 1 in Supplement 1.

The cluster-crossover design was selected to decrease nursing staff workload required to keep track of individual randomization status and to optimize the integration of study procedures into standard ICU workflow. Additional details of the protocol, study design, and statistical analysis approach have been previously published.[22] The trial protocol and statistical analysis plan are available in Supplement 2 and Supplement 3, respectively. Modifications to the original plan included addition of a sixth ICU due to slow patient enrollment and simplification of the statistical model to negative binomial regression analysis. The original plan for advanced biofilm analysis of a convenience sample of CVADs was also unable to be operationalized due to logistics and cost.

Each ICU was randomized to locking all patients with a CVAD in place and at least 1 lumen not in use with 4% t-EDTA (intervention) or control (saline for all catheter types except hemodialysis catheters, which received 4% citrate), reflecting current usual care. Enrollment continued for 3.5 months, after which time each ICU continued delivery of the same locking fluid for 1 additional month to ensure all enrolled patients completed the study under a single treatment condition. Each ICU then crossed over to lock all unused CVAD lumens with the other locking condition for another 3.5 months, followed by a 1-month treatment maintenance period.

All patients admitted to the ICU were screened for eligibility by research staff and enrolled within the first 48 hours of their ICU stay. The hospitals were stratified into 3 pairs based on their study enrollment dates and were randomized within each pair to intervention or control. The randomization schedule was generated by the study methodologist and shared only with the external study manager, who relayed the schedule to the pharmacists responsible for preparing the study locking fluids. The trial methodologist, study manager, and pharmacists were not involved in protocol administration, data collection, outcome adjudication, or primary analysis. Participants, clinical staff members, members of the study team, outcome adjudicators, as well as the primary data analyst remained blinded. Site timelines and treatment sequences are summarized in **Figure 1**.

Figure 1. Flow Diagram of the Trial



Site	Period	
	1	2
1	N = 199 (March-July 2022)	N = 191 (August-November 2022)
2	N = 146 (July-November 2022)	N = 187 (December 2022-March 2023)
3	N = 94 (February-May 2023)	N = 62 (June-October 2023)
4	N = 63 (February-May 2023)	N = 55 (June-October 2023)
5	N = 187 (August-December 2023)	N = 167 (January-May 2024)
6	N = 67 (January-April 2024)	N = 50 (May-September 2024)

CVAD indicates central venous access device. Blue shading represents usual care; purple shading represents 4% tetrasodium EDTA. Numbers of participants enrolled in each cluster period are depicted in each corresponding square, in addition to dates of enrollment. Each cluster period was followed by a 1-month follow-up period during which enrollment ceased in order to maximize the number of patients fulfilling 1 study condition only.

[a]Ineligible due to age, absence of CVAD, no CVAD locking events, or decline to receive blood products. Exact distributions of screening failures were not captured, although anecdotally, the vast majority were due to an absence of CVAD or absence of locking events due to high patient acuity.

[b]Postrandomization exclusions were due to withdrawal of consent (n = 1) and coenrollment in a competing study (n = 1).

All locking procedures were administered by unit nursing staff at the participant level. Any CVADs present during the ICU stay were eligible to receive study locking protocol, including multiple catheters in 1 patient and those inserted outside of the ICU setting. Prior to study start, research staff members provided one-on-one or small-group education sessions to ICU nursing staff outlining the protocol procedures. Staff education documents are included in eAppendix 2 in Supplement 1.

All microbiological testing, medical imaging, and alteplase administration were performed at the discretion of ICU care teams. After a CVAD lumen was no longer needed

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for continuous infusions, an ICU nurse flushed the lumen with 20 mL of 0.9% sodium chloride using a turbulent, push-pause flush method according to standard care. Prefilled, identical, masked study syringes containing 2.5 mL of t-EDTA (KiteLock 4%, SterileCare Inc), 0.9% sodium chloride (for all catheters in the control group except dialysis catheters), or 4% citrate (for dialysis catheters in the control group) were then instilled. Indwelling locking fluid was aspirated whenever possible prior to line reaccess or routine lock exchange. Following locking completion, ICU nurses recorded the lot numbers of each used study syringe. Use of any locking fluid other than the study syringes was designated as a protocol deviation. Research staff maintained a daily presence in the ICUs to ensure adequate supply of study syringes, record protocol adherence and study syringe accountability, and complete outcome data collection. Data were collected and managed using a secure Research Electronic Data Capture (REDCap) database.[23,24] Central line–associated bloodstream infection was centrally assessed in all enrolled patients by the blinded senior investigator (S.R.).

Patients were assessed for all study outcomes throughout their ICU stay until ICU discharge, death, or the end of the treatment maintenance period, whichever occurred first. All outcome events were measured at the participant level, except for administration of alteplase, which was evaluated at the level of the CVAD lumen.

Study Outcomes

The primary outcome was a composite incidence rate of CLABSI, catheter occlusion requiring alteplase use, and/or catheter removal due to occlusion. A CLABSI was defined as a primary bloodstream infection that developed more than 48 hours after CVAD placement without evidence of infection at any other site, in accordance with National Healthcare Safety Network surveillance definitions.[25]

The prespecified secondary outcomes included incidence rates of each of the 3 components of the primary outcome, rates of catheter-associated venous thrombosis, catheter colonization, ICU length of stay, and ICU mortality. Central venous access device colonization was diagnosed in accordance with Infectious Diseases Society of America practice guidelines[26] as growth of greater than 15 colony-forming units from a 5-cm catheter tip segment by semiquantitative culture or growth of greater than 100 colony-forming units from a catheter by quantitative (sonication) broth culture. Diagnostic criteria of remaining outcome measures are provided in eAppendix 3 in Supplement 1. An eighth secondary outcome evaluating costs related to alteplase use for catheter obstruction was planned but never performed due to a decision to include this information in a future formal health economic evaluation.

Statistical Analysis

The planned sample size to achieve 80% power allowing for a $\alpha = .05$ type I error rate was 1524 participants, determined using a simulation assuming a 30% relative reduction in the primary composite outcome rate, using baseline complication rates observed during a 1-month observational period at the senior investigator's site.[2] The 30% relative reduction was considered a clinically relevant and conservative choice given that clinical data available at the time of study planning had shown a 40% to 70% reduction in CVAD complications.[27]

All participants with at least 1 CVAD lock during their ICU stay were included in the intention-to-treat population. Outcome rates were compared using a negative binomial mixed-effects model with a random intercept term for site, catheter-days as the offset term, and adjustment for treatment period, age, sex, compromised immune status, and a combined term for any antibiotic, antifungal, or anticoagulant medication administration at study entry. A sensitivity analysis was performed using the total number of locking events as the offset term to account for variation in locking frequency due to patient acuity. Generalized linear models were fit to assess the effect of intervention on change in calcium over time, with fixed terms for study day 1 calcium value, site, study day, and treatment × study day interaction. Statistical testing was 2-sided with a significance level of $P = .05$. Data summaries were produced using SAS version 9.4 (SAS Institute Inc). All modeling was conducted using the glmmTMB function in R version 4.4.2 (R Foundation).[28]

Results

Trial Population

Between March 22, 2022, and September 25, 2024, 3124 patients were screened, 1574 were enrolled, 1470 received study locking fluid, and 1468 were included in the intention-to-treat analysis (696 receiving t-EDTA and 772 receiving control). Postrandomization exclusions included 1 patient who was enrolled in a competing study and 1 patient who withdrew consent (Figure 1). The baseline characteristics were similar between the 2 groups (**Table 1**), and no participants enrolled during the first randomization period were still hospitalized in the ICU following the crossover. Enrolled patients had a mean age of 60 (SD, 16) years and 37.7% were female. The most commonly used CVADs included triple-lumen central venous catheters (n = 1382 [65.6%]), hemodialysis catheters (n = 218 [10.3%]), trialysis catheters (n = 146 [6.9%]), and triple-lumen peripherally inserted central catheters (n = 122 [5.8%]). Compared with the control group, the t-EDTA group had higher rates of antibiotic use (87.1% vs 79.0%) and systemic or subcutaneous anticoagulant use (68.8% vs 62.4%). Median ICU length of stay was similar between groups (t-EDTA: 6 [IQR, 3-10 days; control: 5 [IQR, 3-10] days). Of 19 993 locking events, there were 10 548 study syringe locks in the control group and 9445 in the t-EDTA group. Additional patient and CVAD characteristics are available in eTables 2 and 3 in Supplement 1.

Primary Outcome

A total of 200 primary outcome events occurred during the study (74 with t-EDTA and 126 with control). Incidence rates of the primary outcome were 13.1 per 1000 catheter-days with t-EDTA vs 19.9 per 1000 catheter-days with control (**Table 2**). Results of adjusted analysis for the primary outcome among patients randomized to t-EDTA vs control yielded a rate ratio (RR) of 0.68 (95% CI, 0.47-0.96; $P = .03$). The adjusted absolute

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Table 1. Baseline and CVAD Characteristics

Characteristics	4% Tetrasodium EDTA (n = 696)	Control (n = 772)
Age, mean (SD), y	60.8 (16.1)	60.5 (16.0)
Sex, No. (%)		
Female	277 (39.8)	277 (35.9)
Male	419 (60.2)	495 (64.1)
Body mass index, mean (SD)[a]	27.8 (7.3) [n = 369]	27.2 (6.6) [n = 401]
Charlson Comorbidity Index, mean (SD)[b]	3.5 (2.7)	3.5 (2.6)
APACHE IV score, mean (SD)[c]	81.5 (31.5)	82.1 (30.8)
SOFA score, mean (SD)[d]	10.1 (3.5) [n = 434]	10.4 (3.7) [n = 463]
Intensive care unit admission, No. (%)		
Nonoperative	530 (76.1)	614 (79.5)
Postoperative	166 (23.9)	158 (20.5)
Immunocompromised, No. (%)[e]	44 (6.3)	50 (6.5)
Central line–associated bloodstream infection prior to study entry, No. (%)	3 (0.4)	8 (1.0)
Occlusion prior to study entry, No. (%)	2 (0.3)	2 (0.3)
Medications		
Systemic antibiotics, No. (%)	606 (87.1)	610 (79.0)
Systemic/subcutaneous anticoagulants, No. (%)	479 (68.8)	482 (62.4)
Systemic antifungals, No. (%)	13 (1.9)	27 (3.5)
Total parenteral nutrition, No. (%)	15 (2.2)	4 (0.5)
CVAD characteristics		
CVAD days, median (IQR)	6 (3-12)	6 (3-12)
Quad-lumen CVC, No. (%)	33 (3.4)	50 (4.4)
Triple-lumen CVC, No. (%)	657 (66.9)	725 (64.4)
Double-lumen CVC, No. (%)	4 (0.4)	6 (0.5)
Single-lumen CVC, No. (%)	2 (0.2)	1 (0.1)
Cordis plus triple-lumen CVC, No. (%)	18 (1.8)	17 (1.5)
Cordis plus double-lumen CVC, No. (%)	3 (0.3)	4 (0.4)
Hemodialysis catheter (double-lumen), No. (%)	97 (9.9)	121 (10.8)
Trialysis line, No. (%)	75 (7.6)	71 (6.3)
Triple-lumen PICC, No. (%)	52 (5.3)	70 (6.2)
Double-lumen PICC, No. (%)	32 (3.3)	51 (4.5)
Single-lumen PICC, No. (%)	6 (0.6)	4 (0.4)
Permacath, No. (%)	3 (0.3)	1(0.1)
Implanted venous access device, No. (%)	0	1 (0.1)
Missing information on CVAD, No. (%)	0	3 (0.3)

Abbreviations: CVAD, central venous access device; CVC, central venous catheter; PICC, peripherally inserted central catheter.

[a] Calculated as weight in kilograms divided by height in meters squared.

[b] The Charlson Comorbidity Index is a scoring system predicting risk of mortality based on presence of select comorbidities (range, 0-6; higher scores denote higher risk).

[c] The Acute Physiology and Chronic Health Evaluation (APACHE IV) score is a severity-of-illness and mortality prediction system based on various physiological parameters present at intensive care unit admission (range, 0-286; higher scores denote higher risk).

[d] The Sequential Organ Failure Assessment (SOFA) score is a systematized scoring system evaluating the extent of organ dysfunction across 6 organ systems (range, 0-24; higher scores denote higher risk).

[e] Immunocompromised was defined as patients who had an organ transplant, were diagnosed with AIDS, had neutropenia (<1000 absolute neutrophils), had corticosteroid administration (>20 mg/d of prednisone or equivalent for more than 6 months), or were currently taking immunosuppressive agents (eg, azathioprine) at the time of enrollment.

rate reduction was 5.1 (95% CI, 0.85-11.19; number needed to treat = 20). Similar rate reduction values were found in the sensitivity analysis (incidence rates, 7.83 vs 11.94 per 1000 catheter-days in the t-EDTA and control groups, respectively; adjusted RR, 0.63; 95% CI, 0.44-0.90; P = .01). Results of the per-protocol analysis are presented in eAppendix 4 in Supplement 1. No data were missing from the primary outcome analysis.

Secondary Outcomes

Among the key secondary outcomes comprising the primary outcome, only CVAD occlusion requiring administration of alteplase significantly differed between the groups. Alteplase use was less common with t-EDTA compared with control (66 vs 112; incidence rates, 11.67 and 17.73 per 1000 catheter-days, respectively; RR, 0.66; 95% CI, 0.46-0.96). There were no significant differences in CLABSI in the t-EDTA group (2 cases; incidence rate, 0.35 per 1000 catheter-days) vs control (3 cases; incidence rate, 0.47 per 1000 catheter-days) (RR, 0.74; 95% CI, 0.10-4.48). Incidence rates of catheter occlusion requiring CVAD removal (1.06 vs 1.74 per 1000 catheter-days; RR, 0.61; 95% CI, 0.21-1.60), catheter-associated venous thrombosis (1.59 vs 0.95 per 1000 catheter-days; RR, 1.65; 95% CI, 0.59-4.94), and colonization (1.95 vs 0.79 per 1000 catheter-days; RR, 2.54; 95% CI, 0.84-8.66) were infrequent, and each was recorded less than 20 times across both groups. There were no differences in ICU mortality (incidence rates, 22.99 per 1000 catheter-days with t-EDTA vs 20.73 per 1000 catheter-days with control; RR, 1.10; 95% CI, 0.86-1.41) (Table 2). Median length of ICU stay was similar between groups (t-EDTA, 6 [IQR, 3-10] days; control, 5 [IQR, 3-10] days). Details of causative organisms of 5 CLABSI and 16 colonization outcomes are listed in

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Table 2. Incidence Rates of Primary and Secondary Outcomes

Outcomes	No. of events		Incidence rate per 1000 catheter-days			
	4% t-EDTA (5654 catheter-days)	Saline or 4% citrate (hemodialysis catheters) (6318 catheter-days)	4% t-EDTA	Saline or 4% citrate (hemodialysis catheters)	Unadjusted difference (95% CI)	Rate ratio (95% CI)
Composite primary outcome						
CVAD-associated bloodstream infection, catheter occlusion requiring alteplase use, and/or CVAD removal due to occlusion	74	126	13.09	19.94	−6.85 (−11.44 to −2.27)	0.68 (0.47-0.96)[a,b]
Secondary outcomes						
Central line–associated bloodstream infection	2	3	0.35	0.47	−0.12 (−0.85 to 0.61)	0.74 (0.1-4.48)[c]
Catheter occlusion requiring alteplase use	66	112	11.67	17.73	−6.05 (−10.38 to −1.73)	0.66 (0.46-0.96)[b]
CVAD removal due to occlusion	6	11	1.06	1.74	−0.68 (−2.01 to 0.65)	0.61 (0.21-1.60)[c]
Thrombosis	9	6	1.59	0.95	0.64 (−0.65 to 1.93)	1.65 (0.59-4.94)[c]
CVAD colonization	11	5	1.95	0.79	1.15 (−0.19 to 2.50)	2.54 (0.84-8.66)[c]
Intensive care unit mortality	130	131	22.99	20.73	2.26 (−3.05 to 7.57)	1.1 (0.86-1.41)[b]

Abbreviations: CVAD, central venous access device; t-EDTA, tetrasodium EDTA.

[a] P = .03; adjusted absolute rate reduction, 5.1; 95% CI, 0.85-11.19; number needed to treat = 20.

[b] Analysis adjusted for treatment period, age, sex, immunocompromised status, and a combined term for any antibiotic, antifungal, or anticoagulant administration prior to study entry, including a random-effect term for site and a fixed-effect term for treatment. Analysis was adjusted for outcomes with greater than 20 events only.

[c] Unadjusted analysis (including only random-effect term for site and fixed-effect term for treatment) performed due to low event counts.

eTable 4 in Supplement 1. Details of length of stay at each participating site are presented in eTable 1 in Supplement 1.

Exploratory Analysis

To determine the impact of adherence to the protocol, we performed a post hoc subgroup analysis of participants for whom at least 80% of CVAD locks were completed using the provided study syringes. Of the 1468 patients included in the intention-to-treat analysis, 1258 (85.7%) maintained at least 80% adherence to the study protocol (n = 613 [88%] with t-EDTA; n = 645 [84%] with control). The primary composite outcome risk reduction in participants who were at least 80% adherent was 0.66 (95% CI, 0.45-0.97) compared with 0.88 (95% CI, 0.38-2.03) in participants who were less than 80% adherent. Site protocol adherence details are presented in eTable 5 in Supplement 1. Complete subgroup analysis is presented in **Figure 2**.

Safety

No adverse events, serious adverse events, or hypocalcemia related to t-EDTA or control locking fluid use were reported during the study period. Ionized or corrected calcium increased moderately over the course of the ICU stay; however, no difference in rate of change over time was observed between treatment groups (β = 0.0004 [SE, 0.01] mmol/L per day; P = .66).

Discussion

In this pragmatic randomized clinical trial of 1468 adult patients in the ICU with CVADs, use of 4% t-EDTA locking fluid reduced the incidence of a composite outcome of CVAD complications compared with control. However, among the 3 secondary outcomes comprising the primary outcome, only the secondary outcome of CVAD occlusion requiring administration of alteplase significantly differed between the groups.

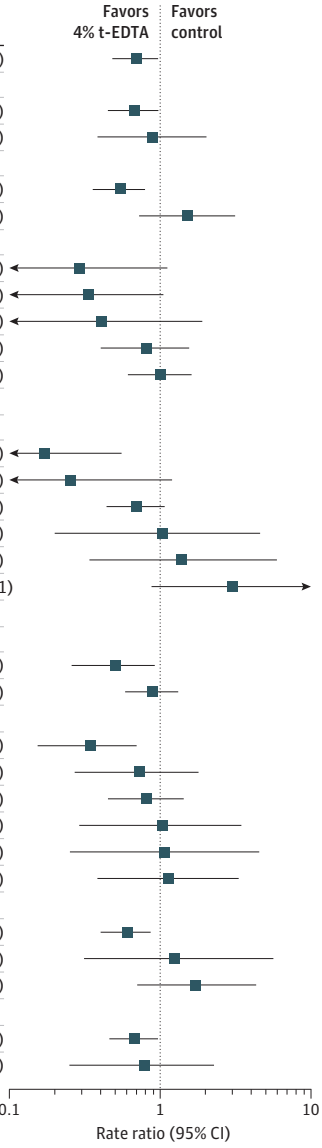
This outcome is biologically plausible owing to potent chelation ability of t-EDTA to reduce intraluminal accumulation of debris caused by early stages of biofilm and thrombus formation, preceding development of CLABSI or symptomatic thrombus. Indeed, all 5 cases of CLABSI observed during the study period occurred in patients with a CVAD duration of at least 7 days, in addition to this subgroup experiencing a greater intervention effect (Figure 2). As the median length of ICU stay during the study period was 6 days across medical-surgical ICUs (and only 3 days in the cardiac surgery center), it is possible that the presence of short-stay patients diluted the treatment effect. Nonetheless, the primary composite outcome occurred significantly more commonly in the t-EDTA group compared with the control group. Furthermore, the observed reduction in alteplase administration is clinically meaningful, as it represents decreased interruptions to clinical care during which nurses spend additional time managing occluded central lines. This is particularly relevant given the current worldwide nursing shortage.[29] Subgroup analysis similarly demonstrates evidence of heightened t-EDTA effect across other high-risk patient profiles, including those with peripherally inserted central catheters and femoral insertion sites.[30]

Two subgroups in this study had unexpected findings. First, patients receiving hemodialysis had no improvement in the primary composite outcome with use of 4% t-EDTA (Figure 2). However, hemodialysis catheters represented approximately 10% of catheters in both study groups, underscoring the need for future adequately powered studies. Second, colonization events were higher in the 4% t-EDTA group vs control (11 vs 5). Absence of standardized protocols regarding

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Figure 2. Dot and Whisker Plot of Overall and Subgroup Analysis of the Primary Outcome

Subgroup	4% t-EDTA		Control		Rate ratio (95% CI)
	No. of events/ No. of days	Incidence rate per 1000 catheter-days	No. of events/ No. of days	Incidence rate per 1000 catheter-days	
Overall	74/5654	13.09	126/6318	19.94	0.68 (0.47-0.96)
Study drug adherence, % of locks					
≥80	59/5045	11.69	91/5184	17.55	0.66 (0.45-0.97)
<80	15/609	24.63	35/1134	30.86	0.88 (0.38-2.03)
CVAD days					
≥7	52/4759	10.93	108/5234	20.63	0.54 (0.36-0.79)
<7	22/2073	10.61	18/2419	7.44	1.50 (0.73-3.14)
CVAD insertion site					
Brachial vein	2/393	5.09	9/507	17.75	0.28 (0.04-1.12)
Basilic vein	3/395	7.59	16/644	24.84	0.32 (0.07-1.05)
Subclavian vein	4/687	5.82	8/637	12.56	0.40 (0.07-1.90)
Femoral vein	19/861	22.07	32/1200	26.67	0.80 (0.40-1.56)
Internal jugular vein	46/4427	10.39	55/4559	12.06	0.99 (0.61-1.62)
Cephalic vein	0/69	0	6/93	64.52	
CVAD type					
Triple-lumen PICC	2/503	3.98	19/770	24.68	0.16 (0.03-0.55)
Double-lumen PICC	2/354	5.65	11/419	26.25	0.24 (0.03-1.20)
Triple-lumen central venous catheter	45/4246	10.6	75/4432	16.92	0.69 (0.44-1.07)
Quad-lumen central venous catheter	3/221	13.57	4/299	13.38	1.01 (0.20-4.60)
Trialysis line	11/571	19.26	12/556	21.58	1.38 (0.34-5.95)
Hemodialysis catheter	10/730	13.7	4/928	4.31	3.04 (0.88-12.21)
Cordis (double/triple)	0/90	0	0/84	0	
CVAD side					
Left	21/1655	12.69	58/2144	27.05	0.49 (0.26-0.92)
Right	53/5177	10.24	68/5496	12.37	0.88 (0.59-1.32)
Site					
Royal Columbian Hospital	13/1881	6.91	36/1906	18.89	0.34 (0.15-0.70)
Nanaimo Regional General Hospital	8/460	17.39	21/761	27.6	0.71 (0.27-1.80)
Surrey Memorial Hospital	30/1349	22.24	48/1667	28.79	0.81 (0.45-1.43)
Burnaby Hospital	10/487	20.53	8/436	18.35	1.02 (0.29-3.46)
St Boniface Hospital	5/1023	4.89	5/1043	4.79	1.06 (0.25-4.54)
Royal Jubilee Hospital	8/454	17.62	8/505	15.84	1.13 (0.38-3.32)
Hemodialysis type					
None	56/4804	11.66	107/5220	20.5	0.59 (0.40-0.87)
Intermittent hemodialysis	6/208	28.85	9/249	36.14	1.23 (0.31-5.64)
Continuous kidney replacement therapy	12/596	20.13	10/810	12.35	1.69 (0.70-4.33)
COVID-19 status					
Negative	65/5170	12.57	114/5785	19.71	0.67 (0.46-0.97)
Positive	9/438	20.55	12/494	24.29	0.78 (0.25-2.28)



CVAD indicates central venous access device; PICC, peripherally inserted central catheter; t-EDTA, tetrasodium EDTA. The primary outcome was a composite of CVAD-associated bloodstream infection, catheter occlusion requiring alteplase use, and/or catheter removal due to occlusion.

catheter removal and microbiological testing and the infrequency of these events (<0.001% of catheter-days) preclude definitive conclusions.

Strengths of this trial include its large sample size, use of triple blinding, inclusion of multiple CVAD types, and evaluation of patients in tertiary and community hospital ICU settings. Use of identical locking and aspiration techniques in both treatment groups decreased the likelihood of behavioral differences in CVAD locking and management. In addition, the study protocol was integrated into existing ICU procedures, increasing external validity of trial results.

Limitations

This study has some limitations. First, the study did not meet the planned sample size of 1524, although 1468 patients (96% of target) were enrolled. Second, admissions to the 5 original participating ICUs initially recruited during the 2020-2022 COVID-19 pandemic years decreased to prepandemic levels mid study, which necessitated recruitment of a sixth ICU. Third, the study was completed over a 2-year period spanning various pandemic-related changes in the critical care environment. It is anticipated that any resulting time bias was adequately addressed by the stratified allocation of

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unit pairs within similar time frames, the cluster-crossover design, and the limited site protocol duration (9 months). Fourth, catheter-related bloodstream infection, defined as a laboratory-confirmed bloodstream infection in which the CVAD is identified as the source of infection through quantitative culture or differential time-to-positivity culture techniques,[26] is a more rigorous outcome measure compared with CLABSI. CLABSI was selected because it does not require systematic CVAD testing and enables more direct comparison of study outcomes with common annual surveillance reporting.[31,32] Fifth, due to logistical and funding constraints, patients enrolled in this study were followed up only during their ICU hospitalization. Sixth, rates of systemic and subcutaneous anticoagulation and antibiotics administered prior to randomization were higher in the t-EDTA group compared with the control group. However, these imbalances were accounted for in the regression model with covariate adjustment. Seventh, the small number of CLABSI and CVAD-associated thrombosis events observed limits interpretability of secondary outcomes and necessitates further evaluation in larger studies.

Conclusions

Compared with a control locking solution, use of 4% t-EDTA locking solution for CVADs reduced the incidence of a composite outcome of CVAD-associated bloodstream infection, catheter occlusion requiring alteplase use, and catheter removal due to occlusion in adult patients in the ICU.

ARTICLE INFORMATION

Accepted for Publication: April 3, 2026.

Published Online: May 18, 2026. doi:10.1001/jama.2026.6025

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Drafting of the manuscript: Ornowska, Blitz, Wong, Reynolds.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: Ornowska, Blitz, Wong, Reynolds.
Obtained funding: Ornowska, Reynolds.
Administrative, technical, or material support: Ornowska, Wittmann, Rohrs, Reynolds.
Supervision: Wong, Rohrs, Reynolds.

Conflict of Interest Disclosures: Ms Wittmann reported receipt of personal fees from Fraser Health Authority for salary support. Dr Vazquez-Grande reported holding an unpaid role as treasurer of the Canadian Critical Care Trials Group and, as such, being a member of its board of directors. Dr Rohrs reported receipt of grants from Fraser Health Authority. No other disclosures were reported.

Funding/Support: Study activities were supported by a grant-in-aid from SterileCare Inc, who also supplied the 4% t-EDTA investigational product. Additional funding was received from the Royal Columbian Hospital Foundation, the CANHealth Network, and the British Columbia Ministry of Health's Health Innovation Pathway Program. Dr Ornowska was supported by a grant from Mitacs (IT184112 and IT31280) as well as the CANTRAIN Doctoral Studentship Award. Dr Reynolds receives salary support from the TB Vets Association.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; or decision to submit the manuscript for publication.

Meeting Presentation: Presented at the International Conference of the American Thoracic Society; May 18, 2026; Orlando, Florida.

Data Sharing Statement: See Supplement 4.

Additional Contributions: We extend our sincere gratitude to all external staff members who made completion of this study possible. In particular, we thank the following individuals for their efforts in study setup, data collection, and protocol administration: Deveshi Deveshi, CRC, Nibi Varghese, CRN, and Chris Condin, PhD, Surrey Memorial Hospital; Fiona Auld, CRN, and Jessica Craighead, CRN, Royal Jubilee Hospital; Linh Vu, CRN, and Sethany Vorng, CRC, Nanaimo Regional General Hospital; Alexander Dapula, CRC, Jonathan Hsiung, CRN, Laurie Figueroa, CRN, and Chantale Pineau, MA, St Boniface Hospital; and Ellie Naseri, CCRP, and Siti Rahmaputri, CCRP, Burnaby Hospital. We also thank dedicated pharmacy staff Marnie Boyle, BSc(Pharm), and Sheena de Castro-Farinas, PA, at St Boniface Hospital, Elaine Tung, BSc, BSc (Pharm), ACPR, Amy Deng, BPharm, MHA, CHE, and Isabella Mathews, PharmD, BSc, ACPR, at Burnaby Hospital, and Vivian Lee, RPh, Deanna Koo, BSc(Pharm), PharmD, RPh, and the entire Calea Pharmacy team. We thank Yongdong Ouyang, PhD, University of Toronto, for his expertise and time in advising on the statistical methodology. We also acknowledge data management support from Irina Chua, BSc(Hons), and Erica Jaaf, BA(Hons), CCDM, at the Centre for Advancing Health Outcomes. All collaborators received compensation for their contributions. Finally, we are sincerely indebted to the dedicated efforts of the critical care nursing staff at Royal Columbian Hospital, Surrey Memorial Hospital, Royal Jubilee Hospital, Nanaimo Regional General Hospital, St Boniface Hospital, and Burnaby Hospital, whose dedication to the study protocol made the study possible.

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